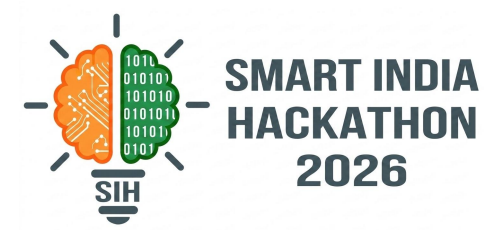


SMART INDIA HACKATHON 2026



KRISHISETU

Problem Statement ID: 26193

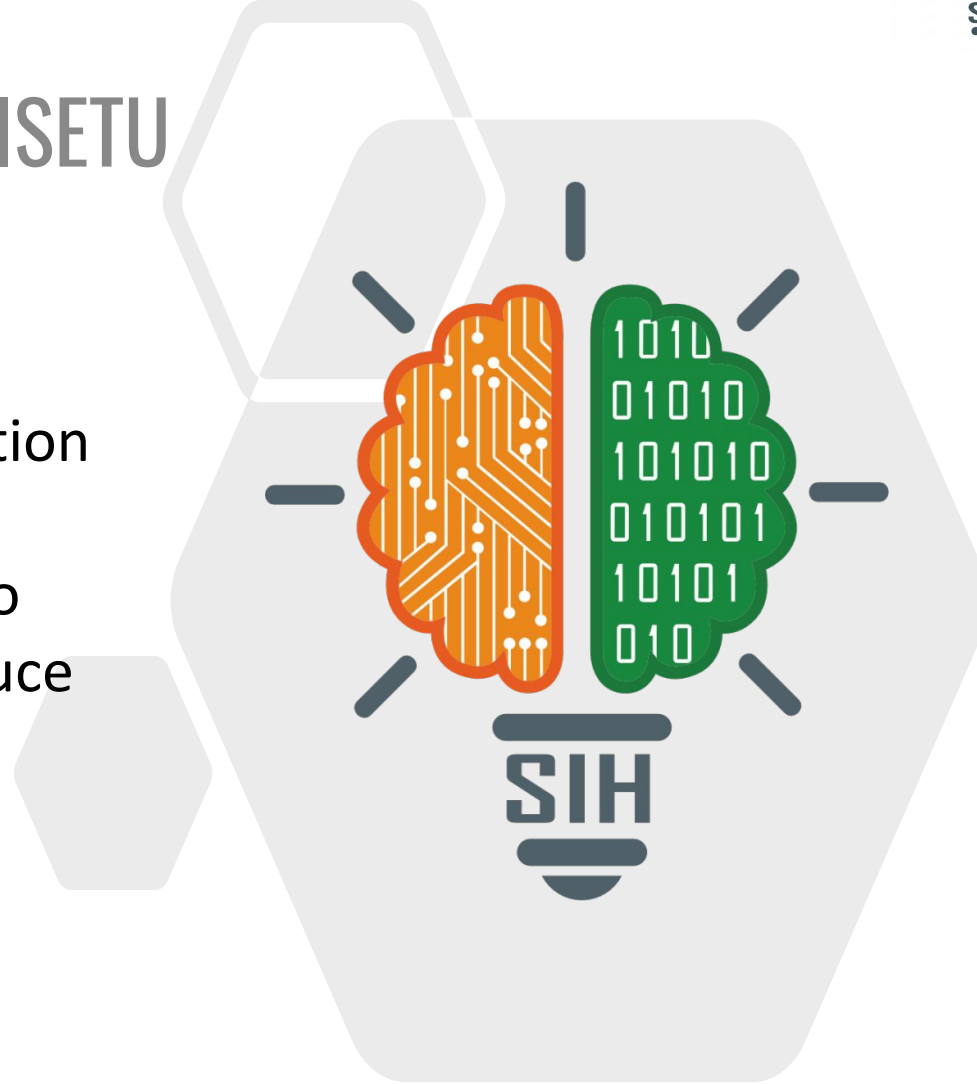
Problem Statement Title: Student Innovation

– Developing solutions to enhance the primary sector of India - Agriculture and to manage and process our agriculture produce

Theme: Agriculture, FoodTech & Rural Development

PS Category: Software

Team Name: LogicForge



Faculty Mentor Approval



Official Approval Email Screenshot

Faculty Mentor Approval – SIH 2026 | Team LogicForge

External

Inbox x



Piyush Rawat
Respected Sir, Good morning Sir. I am Piyush Rawat, Team Leader of Team LogicForge for Smart India Hackathon (SIH) 2026 at Sharda University. Sir, as discussed,

Tue, Sep 1, 2:41 PM (5 days ago)

☆



Dr. Nikhil Sharma (SSCSE Assistant Professor)
to me, 2025266283.manish, 2025508590.vanshika, 2025337654.prabhat, 2025487741.shivani, 2025349734.hridyesh, MANISH ▾
Approved
...
--
Thanks and Regards,

Dr. Nikhil Sharma
Assistant Professor
Sharda School of Computing Science & Engineering
Sharda University, Greater Noida

Ph.D., Department of CSE, Delhi Technology University, Govt. of NCT of Delhi
M.Tech, Department of CSE, NSUT, Govt. of NCT of Delhi
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Wed, Sep 2, 3:02 PM (4 days ago)

☆

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THE CHALLENGES FARMERS FACE



No Data-Driven Guidance

Farmers lack critical insights on post-harvest decisions — when to sell, store, or process.



Market Blind Spots

Limited visibility into real-time market prices, localized demand signals, and shelf-life estimates.



Avoidable Post-Harvest Losses

Heavy financial losses due to poor timing, suboptimal decision-making, and disconnected buyers.



Fragmented Ecosystems

Existing legacy systems tackle only isolated tasks (price prediction OR disease detection) but are never unified.

OUR SYSTEMATIC INTUITIVE SOLUTION



SELL NOW

Optimal market timing recommendations paired with reliable confidence scores.



STORE

Shelf-life forecasting and quality-driven smart storage guidance.



PROCESS

Seamlessly route excess produce to value-added processing units.



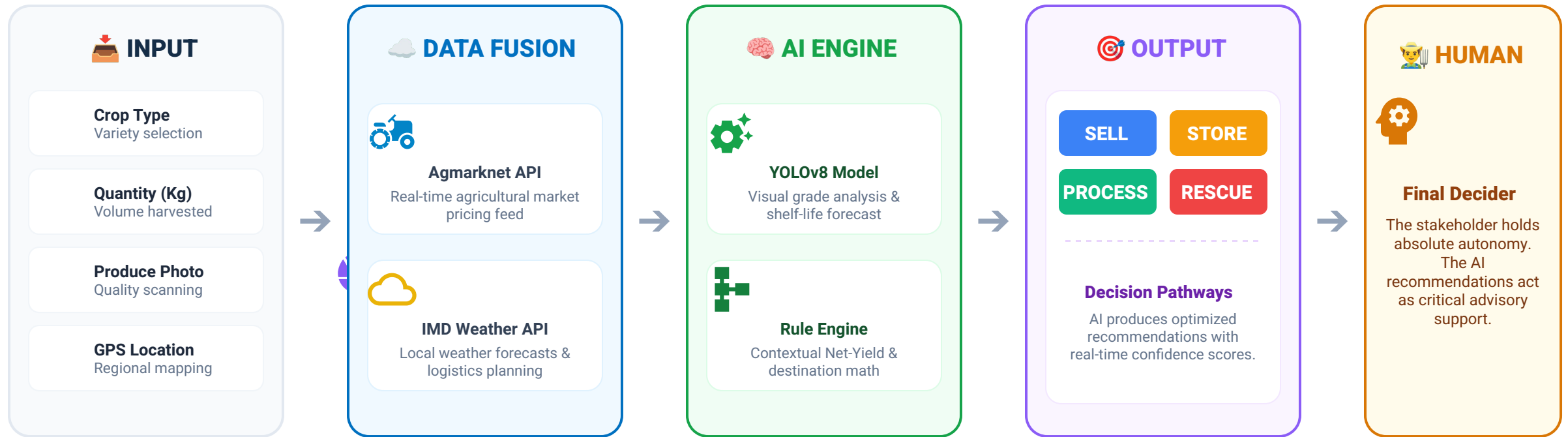
RESCUE

Directly connect with alternative bulk buyers before spoilage occurs.



Human-in-the-Loop Paradigm

AI provides advanced recommendations, but the final decision always rests with the farmers.



⚡ TECH STACK

Python | FastAPI | PostgreSQL | TensorFlow.js | React | AWS Cloud Infrastructure



Risk Assessment & Feasibility Matrix

Challenge	Mitigation Strategy	Complexity	Status
Data Quality	Validation & preprocessing pipelines before CV analysis	MEDIUM	✓
Market Volatility	Introduce reliable confidence scores instead of absolute predictions	HIGH	✓
Regional Variation	Deploy localized, crop & region-specific model variants	LOW	✓
Quality Estimation	Future integration of cost-effective IoT sensors & image analysis	MEDIUM	✓
Connectivity Constraints	Build offline-friendly, highly compressed low-bandwidth interface	LOW	✓
Model Drift	Implement continuous automated retraining utilizing direct farmer feedback loops	MEDIUM	✓

WHY IT SCALES



Cloud Deployable

Cloud-first architecture for rapid, seamless wide-scale deployments.



Modular CV Architecture

Models improve independently without affecting core pipeline.



Expandable Crop-by-Crop

Scales iteratively from tomato models to diverse alternative cash crop varieties.



Ecosystem Impact

For Farmers: Realize data-driven post-harvest decisions to slash avoidable spoilage losses and optimize market timing.

For Buyers & Processors: Gain transparent regional produce visibility, simplifying high-efficiency sourcing of specific quality grades.

For the Ecosystem: Create streamlined produce routing to minimize aggregate food waste and maximize logistics resources.

Future Scope

Multilingual Voice Interaction: Enable farmers to seamlessly interact with LogicForge using regional voice commands.

IoT Cold Storage Monitoring: Direct sensor integrations to update perishables' shelf-life predictions in real-time.

Real-time Market Bidding: Facilitate instant, friction-free matching and transaction completion directly in-app.

Explainable AI (XAI): Deliver clear, visual explanations for every decision and storage recommendation.

Designed to Scale

Crop Expansion: Broaden current models from Tomatoes to other perishables, scaling eventually to grains.

Geographical Growth: Roll out iteratively from district levels to state-wide and nationwide implementations.

Linguistic Accessibility: Deploy continuous localization and multi-language support to empower diverse regions.

Ecosystem Onboarding: Expand the partner network with cold storages, logistics entities, and processing units.

FARMERS • FPOs • AGGREGATORS • TRADERS • PROCESSORS • BUYERS • COLD STORAGES • LOGISTICS PARTNERS

Government Data

15%

Total Post-Harvest Loss
Reported on key agricultural commodities

ICAR & NABCONS 2022

The landmark joint study assessing harvest and post-harvest losses of major agricultural produce in India.

Key Insights:

- Validates the massive 15% physical produce loss problem across grains, fruits, and vegetables.
- Pinpoints market delays, inadequate cold chain storage, and uncoordinated supply routing as primary drivers.
- Directly establishes the scale of economic recovery possible with optimized Sell/Store decision intelligence.

Research Papers



IEEE Publication
Computer Vision Framework

Computer Vision for Product Quality

Academic framework validating image-based automated crop grading and shelf-life estimation.

Technical Validation:

- Backs our YOLOv8 approach for real-time visual quality and disease detection on device cameras.
- Proves accuracy parity between deep learning-driven surface feature analysis and expert manual inspection.
- Serves as the core methodology backing our automated shelf-life degradation algorithms.

Competitor Matrix

Capabilities	Legacy	LogicForge
Price Forecasting	✓	✓
Quality Grading	✗	✓
Storage Decisions	✗	✓
Processing Routing	✗	✓
Unified Decision Engine Gap	Legacy players offer isolated utilities. LogicForge integrates parameters to suggest the absolute next best step.	